
INVESTIGATING MULTIMODAL TRANSFER LEARNING APPROACHES TO SKIN LESION CLASSIFICATION

TECHNICAL REPORT

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<https://github.com/>

ABSTRACT

Skin lesion classification remains challenging due to limited data, features, class imbalance, and the underutilization of patient metadata. This work investigates multimodal transfer learning approaches that integrate metadata - age, sex, and lesion location - into a frozen ResNet50 backbone. We compare two strategies: mid-level feature fusion and Feature-Wise Linear Modulation (FiLM), which adaptively conditions intermediate activations on metadata. Experiments on the HAM10000 dataset demonstrate consistent performance gains over image-only baselines, with up to 8.7% improvement in weighted F1-score. Deeper FiLM insertion points yield stronger feature conditioning but also amplify bias toward metadata-dominant classes. These findings reveal that while deeper metadata integration can improve overall performance, it may also amplify bias toward overrepresented classes, illustrating a key trade-off in multimodal medical image classification.

Keywords Transfer Learning (TL) · Balanced Sampling · Data Augmentation · Test-Time Augmentation (TTA) · Feature-Wise Linear Modulation (FiLM) · Metadata Fusion · Learning Rate Scheduling · Fine-tuning · Convolutional Neural Networks (CNN)

1 Introduction

Skin cancer is one of the most common and deadly forms of cancer worldwide, with over 60,000 recorded deaths in 2022^[1]. Early detection and accurate diagnosis are critical for improving patient outcomes, but manual diagnosis is time-consuming and highly dependent on clinical expertise. In recent years, deep learning has demonstrated remarkable success in medical image classification tasks, offering automated support to dermatologists through computer-aided diagnostic systems.

Despite this progress, deep neural networks typically require large annotated datasets and may not generalize well to new patient populations or imaging conditions. Transfer learning (TL) has become a popular strategy to address these challenges by leveraging pre-trained models on large-scale datasets and fine-tuning them on domain-specific medical images.

Most existing approaches rely solely on image data and ignore valuable contextual information (metadata) such as patient age, gender, and lesion location - all of which can contribute to a more accurate prediction. Recent work has shown that conditioning neural networks on auxiliary information can improve performance in multimodal tasks. Feature-wise Linear Modulation (FiLM) is one promising technique for integrating metadata into convolutional architectures.

<https://www.iarc.who.int/cancer-type/skin-cancer/> (Accessed 24/10/2025)

In this study, we investigate the effectiveness of incorporating metadata using traditional fusion of metadata in comparison to FiLM layers, within a transfer learning framework for skin lesion classification. We evaluate how the inclusion of patient metadata influences model accuracy and robustness compared to standard transfer learning baselines. Our findings aim to provide insights into the potential of multimodal architectures for improving diagnostic reliability in automated dermatological analysis.

1.1 Research Question

Research Question

Can the integration of metadata using Feature-wise Linear Modulation (FiLM) improve the diagnostic performance and generalization of transfer learning models for skin lesion classification?

1.2 Methodological Overview

We utilise a pre-trained ResNet50 convolutional neural network in conjunction with the HAM10000 dataset, developing and thereafter comparing a standard baseline TL model with variants that incorporate patient metadata. Classification accuracy, precision, recall, F1/F2-score are used to assess performance.

2 Related Works

2.1 HAM10000

The HAM10000 dataset was released in 2018 with the purpose of serving as a training set for academic machine learning purposes, and has since then served as a comprehensive resource for training and evaluating machine learning models in dermatological diagnostics. The images were collected over a period of 20 years, with the process including a semi-automated workflow and manual reviews, ensuring consistent colour and luminance. Labels were confirmed by pathology; the remainder are validated through follow-up, expert consensus, or in-vivo confocal microscopy [1]. The dataset hence serves as an ideal medium to test our research question.

2.2 Transfer Learning

Deep learning introduces challenges with temporal, computational, and data resources needed to train large models. Transfer learning aims to combat this by taking the capabilities of a model trained on one task, and bridging it to a new related task. Transfer learning hence provides an efficient way of learning when resources are limited [2]. In the world of deep learning for image classification, this can be done by utilising a pre-trained encoder, and supplementing this with a trainable decoder. This technique has proven successful in skin lesion classification, with studies achieving over 90% accuracy in many cases [3][4][5]. ResNet50 has proven particularly effective due to its performance and efficiency, especially with medical imaging tasks [5].

2.3 Multimodal Learning

Multimodal deep learning entails using different input types to enhance learning. In the case of skin lesion classification, metadata has been successfully incorporated into image classification models to improve performance. Performance has been proven to increase between baseline models and metadata fusion models, with [6] citing performance increases in all their trialled fusion methods, and [7] citing an accuracy increase of 6.4% using a ResNet50 backbone.

[7] further conducted a systematic comparison of multiple metadata fusion techniques and found that mid-level fusion, where patient metadata are concatenated with the CNN’s internal feature activations (e.g., from the fully connected layers), yielded the highest overall performance. The internal activations capture high-level visual abstractions, and combining them with metadata enables the model to learn joint correlations between clinical and visual cues. However, while this approach improved overall accuracy, it also tended to reduce per-class sensitivity, indicating that minority lesion types benefited less from metadata integration.

Building on these findings, our work adopts a similar mid-level fusion strategy for comparison, while additionally introducing Feature-wise Linear Modulation (FiLM) as a deeper, more adaptive conditioning mechanism for metadata integration.

2.4 FiLM

FiLM was introduced in 2018 as a state-of-the-art method across a variety of visual reasoning tasks. It introduced feature-wise linear modulation layers, which condition neural network activations by applying learned, input-dependent affine transformations:

$$\text{FiLM}(F_{i,c} \mid \gamma_{i,c}, \beta_{i,c}) = \gamma_{i,c} \cdot F_{i,c} + \beta_{i,c} \quad (1)$$

The original experiments were focused on how linguistic input could affect the CNN interpretation of an image, with questions being asked about an image, and an answer being outputted [8]. Since then, the technology has been adapted, in particular with the incorporation of metadata in deep learning frameworks, which has proven that early metadata integration could substantially improve results [9] [10]. As of writing, there are no indications that FiLM has been tested in a skin lesion classification context.

3 Methods

3.1 Test setup

The experiments were conducted on a MacBook Pro M4 Pro (16-core CPU, 20-core GPU, 24 GB RAM) using PyTorch, which provided full control over model configuration and automated generation of performance metrics and visualizations. Configurations were implemented as modular config classes, defining all adjustable parameters related to the experiment, model, and data augmentations. Each experiment was executed by selecting a configuration class, which supplied the parameters to the corresponding training and evaluation methods. The dataset was split into 80 % training and 20 % validation data. Due to time and hardware constraints, all experiments were performed on 30 % of the dataset for 20 epochs, a balance determined through preliminary testing. Input images were resized from 28x28 to 224x224 (ResNet50 required input size) using bilinear interpolation due to the same aforementioned constraints hindering optimal workflow on the full resolution images.

3.2 Dataset

The HAM10000 dataset, comprising 10,015 samples distributed across seven distinct lesion classes. Each sample has associated metadata, relating to the patients' age, sex, and lesion localisation. There are two main observations relevant to the dataset, described below.

3.2.1 Class Distribution & Imbalance

The dataset is characterized by a significant class imbalance.

- The most common class, *Melanocytic Nevi (nv)*, accounts for the vast majority of samples with 6,705 (66.9 %).
- The four smallest classes - *akiec*, *bcc*, *df* and *vasc* - collectively represent only about 11.1 % of the total samples.

The extent of the imbalance suggests significant problems. With lack of relevant samples for the classes of low representation, the model may struggle to learn their relevant features. The experiments described in section 3.3 highlight the problems and our proposed solution.

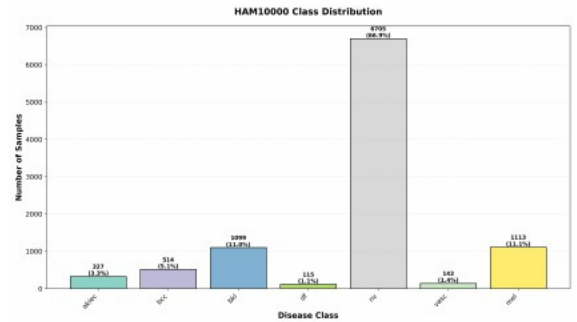


Figure 1: Class distribution of the HAM10000 dataset.

3.2.2 Metadata Distribution Across Classes

The metadata features are distributed unevenly across the disease classes, suggesting their potential utility for classification (See figure 2).

The age distribution by class shows that the malignant classes like *akiec* and *bcc* tend to have a higher median age (around 70-75 years), while the most common class *nv*, shows a lower median age (around 45 years).

The sex distribution reveals that *melanoma* (*mel*) samples show a clear bias toward male patients, while other classes, including *nv* are more balanced, but still slightly male-dominant.

The back and lower extremity are the most frequent locations overall. Notably, *nv* samples are heavily concentrated on the back, trunk, and lower extremity. The *df* furthermore has a clear correlation, as the associated samples are likely to be located on the lower extremity.

These correlations between class and metadata features, indicate that patient-specific data can provide relevant, non-visual cues. This supports the hypothesis that incorporating metadata can potentially improve classification performance.

3.3 Experiments

Seven iterations were conducted before convergence on a stable baseline suitable for metadata-related experiments. Each of the 7 iterations consisted of several sub-iterations involving targeted hyperparameter tuning. Both macro-average and weighted average was tracked, however macro-average is preferred due to the medical criticality and biased nature of the data, i.e.

$$\text{Macro Average} = \frac{1}{N} \sum_{i=1}^N \text{Metric}_i$$

Table 1: Macro Average Performance Metrics of Different Model Configurations

Configuration	Accuracy	Precision	Recall	F1	F2
Baseline (SGD)	0.675	0.217	0.188	0.179	0.183
Baseline (ADAM)	0.720	0.446	0.396	0.393	0.391
+ Balanced Sampling	0.657	0.470	0.523	0.480	0.500
+ Aug. & TTA	0.655	0.499	0.649	0.540	0.593
+ Advanced Head	0.693	0.523	0.638	0.557	0.597
+ LR Scheduler	0.690	0.553	0.658	0.582	0.620
- Fine-tuning	0.693	0.541	0.665	0.578	0.622

The initial baseline consists of a standard SGD optimizer, with a cross-entropy loss function and a single linear layer as the classification head, achieving a macro F1-score of only **0.179** (Table 1). Inspection of the confusion matrix (Figure 3) revealed that class imbalance was a significant issue, with clear overfitting toward majority classes. Under class imbalance, samples from minority classes appear less frequently in each mini-batch. As a result, the gradients $g_{t,i}$ associated with these classes are updated less often and with higher variance. Since all parameters share the same learning rate α , the effective updates for parameters tied to minority classes become inconsistent and noisy - leading to slower learning and poorer performance for those classes [11][p. 4]. Switching to the Adam optimizer improved F1 to **0.393**, reflecting its adaptive learning rate mechanism $\frac{\alpha}{\sqrt{v_{t,i} + \epsilon}}$, which scales updates more effectively for infrequently updated parameters. Introducing balanced oversampling

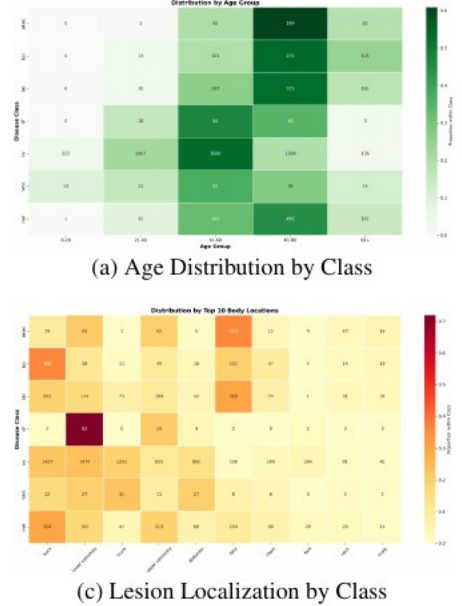


Figure 2: Key metadata distributions across disease classes.

we chose to treat each class equally:

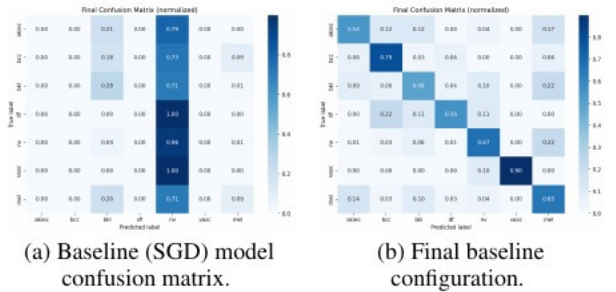


Figure 3: Comparison of baseline (a) and improved (b) model performance.

(i.e. 21 samples per batch $\rightarrow$ 3 per class) further improved recall from **0.396** to **0.523**, demonstrating that equal class representation in batches allows minority classes to contribute more consistently to gradient updates. Adding data augmentation and TTA increased recall substantially to **0.649** and F2 to **0.593**, confirming the synthetic diversity mitigates overfitting and prediction averaging mimics the noise introduced by augmentation (Figure 4). It became evident in our training sessions, that the learning capacity of the model was too low by the way training accuracy began stagnating at 66-67 %.

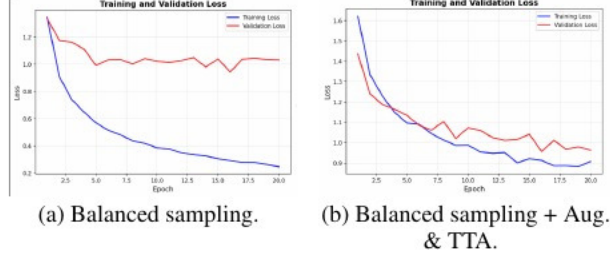


Figure 4: Validation loss comparison between balanced sampling and augmentation with TTA.

The former experiment motivated the introduction of a more advanced head (2), which would be able to enhance the model’s ability to capture non-linear feature interactions and improve class separability. While the pre-trained ResNet50 provides rich, high-level representations, a single linear layer may not fully exploit these features for complex, multi-class distinctions such as those found in skin lesion classification. The new head adds moderate depth and regularization to promote better generalization without overfitting, yielding better results in most metrics.

Formally, if the pre-trained encoder outputs feature vector $h \in \mathbb{R}^{2048}$, the advanced head applies successive transformations:

$$\begin{aligned} z_1 &= \text{ReLU}(\text{BN}(W_1 h)), & W_1 &\in \mathbb{R}^{256 \times 2048}, \\ z_2 &= \text{Dropout}(z_1, p = 0.3), \\ \hat{y} &= W_2 z_2, & W_2 &\in \mathbb{R}^{7 \times 256}. \end{aligned} \quad (2)$$

First linear layer projects the high-dimensional feature vector from ResNet50 into a more compact, learnable representation, reducing dimensionality while retaining discriminative information. **BN** normalizes activations to stabilize and accelerate training, improving convergence and reducing internal covariate shift. **ReLU** introduces nonlinearity, enabling the network to model complex relationships between features and classes. **Dropout** deactivates 30 % of neurons during training to prevent overfitting and encourage robust feature learning. Lastly there is a mapping between the learned representations to the seven output classes, producing the final class logits for prediction.

Building on the introduction of the advanced head, which improved non-linear feature interactions and class separability, we observed that the model could quickly reach promising regions of the loss landscape. However, continuing training with a fixed or overly aggressive learning rate risked overshooting these minima, limiting further gains. Implementing a learning rate scheduler allows the model to explore broadly at the start and gradually refine its search, stabilizing convergence and enhancing generalization. This approach complements the advanced head, contributing to the observed improvements in most metrics, particularly those reflecting minority-class performance and overall robustness, as shown in Table 1.

Despite the potential benefits of fine-tuning deeper layers, we chose not to include it in our final baseline model. Fine-tuning introduces a significantly higher degree of sensitivity to hyperparameter choices, such as learning rate ratios between the head and backbone, unfreezing schedules, and regularization strength. In preliminary experiments, while fine-tuning occasionally yielded small improvements in validation metrics, it also led to less stable convergence and greater variance across runs – particularly given the limited data and severe class imbalance in HAM10000. Moreover, since our objective for the baseline was to establish a robust and reproducible reference point rather than maximize performance, we prioritized stability and generalization under a fixed backbone. This allows the baseline in a later stage to more clearly isolate the effects of FiLM and metadata fusion, without the added variability introduced by fine-tuning dynamics.

3.4 Architecture

The described experiments enabled a conclusion of a final model architecture to test our research question. Figure 5 visualises the overall model structure with simultaneous example setups for FiLM and Fusion. ResNet50 serves as the base model for transfer learning where we implemented and replaced the final layer with a complex classifier head, elaborated in equation (2). The white layers and their connections serve as the two implementations we wish to compare. The FiLM layer receives the transformation variables from the generator, which are created from the given metadata. The layers in the generator have weights that are trained in the same way as the head, such that the model can transform the outputs from the second layer effectively. The "concat" layer serves as the implementation of metadata fusion, where the processed metadata (which includes normalising and one-hot encoding the values), is concatenated

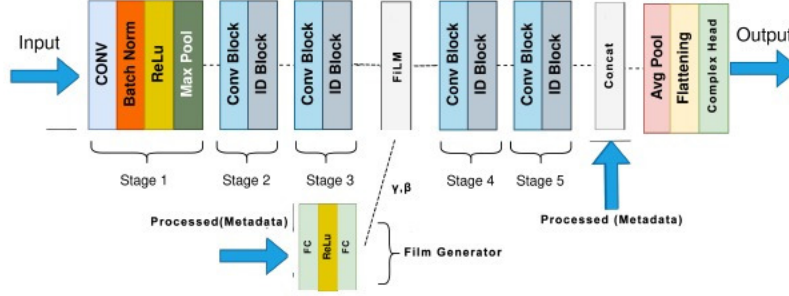


Figure 5: Architecture Visualisation

with the outputs from the final layer before the new head. The overview of the model features as they were added is given in Table 1 and the exact experiments conducted to arrive at the architecture can be seen in the source code.

4 Results

In this section, we present the empirical results of our investigation into multimodal transfer learning approaches for skin lesion classification. We detail the performance of models incorporating metadata through FiLM at various depths within the network, as well as a concatenation-based metadata fusion approach. Our findings are assessed using accuracy, precision, recall, F1-score, and F2-score, and are subsequently compared against the established baselines and relevant literature.

4.1 Overall Performance

When establishing a robust and well-performing baseline model, we valued macro-average. However due to previous papers on metadata incorporation, and evidence of a minority-class worsening, we include weighted averages as well, allowing for comparisons.

$$\text{Weighted Average} = \frac{\sum_{i=1}^N w_i \cdot \text{Metric}_i}{\sum_i 1^N w_i}, w_i = \text{support for class } i$$

Table 2: Macro/Weighted Avg. Performance Metrics across Insertion Points & Fusion

Configuration	Accuracy	Precision	Recall	F1	F2
Baseline	0.690	0.553 / 0.768	0.658 / 0.653	0.582 / 0.686	0.620 / 0.673
FiLM Stage 3	0.687	0.486 / 0.771	0.624 / 0.663	0.530 / 0.696	0.578 / 0.682
FiLM Stage 4	0.713	0.562 / 0.780	0.591 / 0.712	0.557 / 0.733	0.572 / 0.725
FiLM Stage 5	0.728	0.555 / 0.792	0.645 / 0.718	0.587 / 0.742	0.618 / 0.732
Metadata Fusion	0.740	0.609 / 0.796	0.682 / 0.723	0.631 / 0.746	0.657 / 0.737

4.2 Comparison with Related Works

Our experiments as presented in Table 2 confirm prior findings from [6] and [7] that incorporate patient metadata consistently improves overall model performance compared to image-only baselines. In particular, both FiLM-based conditioning and mid-level fusion yielded higher weighted-average metrics, demonstrating that multimodal integration can enhance diagnostic robustness, however at a cost of minority class representation.

The magnitude of improvement in our experiments $\approx 5.5/8.2\%$ and $\approx 7.2/8.7\%$ for FiLM (stage 5) and fusion respectively across accuracy and weighted F_1 for fusion is comparable to that reported by [7], suggesting metadata continues to provide complementary diagnostic cues even when integrated through distinct mechanisms such as FiLM.

While sources [9][10] indicated state of the art performance with FiLM incorporation, metadata fusion consistently outperformed FiLM across all metrics. This is likely because the fusion mechanism directly concatenates the processed metadata with high-level visual features at the same network stage (Stage 5), allowing the classifier head

to jointly learn correlations without relying on the intermediate learning stages required by FiLM. FiLM’s success depends on learning optimal parameters, which may require longer training or more balanced data. Consequently, under the experiments constraints, mid-level fusion proved more effective for integrating metadata.

4.3 Comparison of Insertion Points

The placement of the FiLM layers within the ResNet50 backbone proved to be a critical factor in model performance, with deeper insertion points yielding better results. Deeper layers in a CNN tend to learn more complex and abstract features, meaning FiLM insertion at a later stage applies the metadata driven modulation to richer more semantically meaningful features. This in turn could be the reason for more informed diagnostics.

Table 3: Trainable parameter counts: classifier head vs. FiLM generators at different insertion depths.

Component	Channels C	Extra params	% of ResNet50	Total params
Classifier head (fc)	–	–	–	526,855
FiLM generator (Stage 3, layer2)	512	134,400	0.526%	661,255
FiLM generator (Stage 4, layer3)	1024	266,496	1.043%	793,351
FiLM generator (Stage 5, layer4)	2048	530,688	2.077%	1,057,543

Table 3 summarizes the additional parameters introduced by the FiLM generators. Because each generator scales linearly with the number of feature channels ($2C$ outputs for γ and β), deeper insertion points incur larger parameter counts. Even at Stage 5, this represents only $\sim 2\%$ of the total ResNet50 parameters, suggesting that performance differences arise primarily from representational effects rather than model capacity. With the backbone frozen, larger FiLM modules mainly increase the model’s sensitivity to metadata correlations - enhancing weighted metrics driven by majority classes, but not necessarily improving macro-level balance across all lesion types.

4.4 Comparison with Metadata Distribution

In the case of metadata incorporation, the model gains the ability to map metadata features to internal feature transformations. However, because both the dataset and metadata are class-imbalanced and correlated, this flexibility can lead the network to overfit to spurious correlations rather than learn more general multimodal patterns. The correlation becomes evident when comparing Figure 6 to Figure 2. The model tends to misclassify lesions whose metadata distributions, particularly age (seemingly the most useful [7]), are highly correlated. For instance, *df* is frequently confused with *bcc*, which aligns with their overlapping dominant age ranges. In contrast, *vasc* is more accurately distinguished, likely due to its relatively uniform age distribution, which provides weaker correlation cues for the model to exploit.

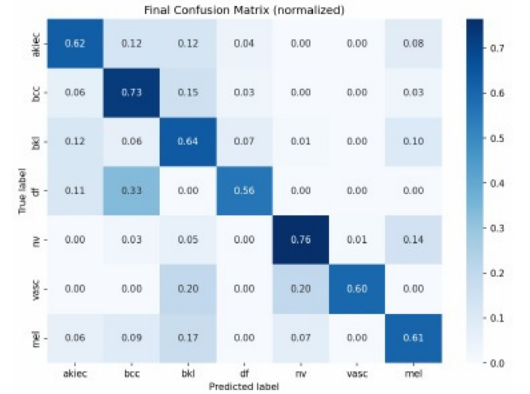


Figure 6: Confusion matrix for FiLM Stage 5 showing correlation bias toward age distribution.

5 Discussion & Future Work

5.1 Classification & Metrics

Our study addressed a multiclass classification task across seven lesion categories. While this setting mirrors real clinical complexity, the severe class imbalance and inter-class variability introduce substantial challenges. Future work could instead evaluate the multimodal approaches in a binary setting (malignant vs. benign), where clearer class boundaries may better reveal the benefits of FiLM and concatenation. Simplifying the task could thus expose stronger diagnostic performance and more accurately reflect the potential of metadata integration.

Another important consideration is the choice of evaluation metrics. Although overall performance was primarily assessed using the F1 score, recall for malignant classes may provide a more meaningful measure of effectiveness. In practical applications, the ability to correctly identify malignant cases is critical, as failing to detect them has far greater consequences than false positives. Thus, recall better reflects real-world performance in this context.

5.2 Metadata Augmentation

Our experiments revealed that while metadata integration generally improved overall performance, it also appeared to amplify bias towards the overrepresented classes. This effect suggests that the model may have learned certain metadata patterns and their relationship to the larger classes. To mitigate this, we believe something like metadata augmentation could serve as a plausible solution. We hence suggest controlled perturbations on the metadata values, such that the model may become more robust to the metadata-induced biases and better generalize across minority classes, as we saw happen when introducing the technique on the images themselves.

5.3 Computational & Temporal Limitations

As mentioned, all experiments were conducted on a fraction of the data and for a constant 20 epochs. While this setup enabled consistent experimentation within practical limits, it inevitably restricted the model’s absolute learning capacity. However, as all models operated under identical constraints, the relative performance differences remain a valid basis for comparing the integration strategies. We believe both the FiLM and concatenation models likely underperformed relative to their potential on the full dataset, as very little data, especially for the under-represented classes, were available to learn from. Similarly, restricting training to 20 epochs prevented some of the models, especially the more complex FiLM variant, from fully converging, biasing performance in favour of architectures that learn faster but generalize less effectively.

Furthermore, the employment of balanced oversampling to address class imbalance likely constrained performance within this reduced subset. By oversampling minority classes on only 30% of the data, the model was repeatedly exposed to a very small pool of unique examples, increasing the risk of overfitting to those specific instances. However, we retained this strategy (as opposed to undersampling) as we hypothesize its effectiveness would be fully realized on the complete dataset, where a larger absolute number of minority samples would allow for robust class balancing without the same degree of repetition.

6 Conclusion

This study investigated the integration of patient metadata into transfer learning models for skin lesion classification, comparing multimodal models such as FiLM and mid-level fusion to baselines. The results demonstrated that incorporating metadata consistently improved diagnostic robustness compared to image-only methods. Among the approaches, metadata fusion yielded the best overall results, likely due to its direct integration of attributes into high level feature representations. However the incorporation of metadata also amplified bias towards majority classes, and the testing conditions perhaps obscured the ceiling of fusion and particularly FiLM. On this basis, future work should build on the findings and structure of this exploration, and investigate the effect of different classification methods, performance metrics, and metadata handling within a more robust computational and temporal framework. Ultimately, the work reinforces the value of multimodal deep learning in dermatological diagnostics, and highlights the importance metadata can play in such models.

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Declaration of use of GAI tools

We have used the following generative artificial intelligence (GAI) to complete this project:

- Gemini 2.5/3 pro

We used GAI tools in the following way:

- Support with Python and Pytorch syntax.
- Support with chart generation.
- Assisted in refining text for conciseness and clarity.